MIL-C-915/38B
5 April 1973
SUPERSEDING 1/
MIL-C-915/38A
1 August 1972

MILITARY SPECIFICATION

CABLE, ELECTRICAL, TYPES TCJX, TCKX AND TCTX

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (Watertight)

First - Stranded conductor, uncoated (see details for size and stranding).
Second - Extruded silicone rubber insulation (see details for dimensions).
Third - Glass braid, color identified (see color identification code table contained herein) (See details for dimensions).
(B) Fourth - Two conductors cabled to form a pair, with lay not greater than 2.25 inches. For TCJX, one iron and one constantan conductor. For TCKX, one chromel and one alumel conductor. For TCTX, one copper and one constantan conductor.
Fifth - The specified number of pairs (see details) cabled together with a lay not greater than 10 inches. Either non-fibrous or a combination of non-fibrous and asbestos fillers shall be used to form a firm, round assembly. (Non-fibrous filler shall be silicone.)
Sixth - Binder tape applied helically with overlap.
Seventh - Silicone rubber jacket (see details for thickness).
Eighth - Braided metal armor.
Ninth - Paint.

Details

Cable type and size	Conductors: No. of prs.	Conductors: Number of strands and strand dia. (inch)	Conductors: Dia. over insul. min. (inch)	Conductors: Dia. over braid max. (inch)	Cable jacket thick. min. (inch)	Overall dia. max. (inches)	Cold bending mandrel diameter maximum (inches)
TCJX-3	3	7/0.0201	0.100	0.125	0.040	0.742	9.0
TCJX-7	7	7/0.0201	.100	.125	.050	.983	11.0
TCJX-12	12	7/0.0201	.100	.125	.050	1.269	15.0
TCKX-1	1	7/0.0201	.100	.125	.030	.456	5.0
TCKX-3	3	7/0.0201	.100	.125	.040	.742	9.0
TCKX-7	7	7/0.0201	.100	.125	.050	.983	11.0
TCKX-12	12	7/0.0201	.100	.125	.050	1.269	15.0
(B) TCTX-1	1	7/0.0113	.065	.085	.030	.350	4.25
TCTX-3	3	7/0.0113	.065	.085	.030	.552	6.5
TCTX-7	7	7/0.0113	.065	.085	.040	.731	8.0
TCTX-12	12	7/0.0113	.065	.085	.050	.964	11.0

1/ Requirements formerly covered on MIL-C-23206/1.

(B) Denotes changes

FSC 6145

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Color identification code

Pair No.	Constantan or alumel	Iron, copper or chromel	
		Base color	Tracer color
1	Red	White	- - -
2	Red	Black	- - -
3	Red	Green	- - -
4	Red	Orange	- - -
5	Red	Blue	- - -
6	Red	White	Black
7	Red	White	Green
8	Red	White	Orange
9	Red	White	Blue
10	Red	White	Red
11	Red	Black	White
12	Red	Green	White

EXAMINATION AND TESTS:

	Requirements:
Basic Electrical:	
Conductor resistance - Ohms/100 ft. loop, maximum	14.41 for TCJX 24.53 for TCKX 47.08 for TCTX
Voltage withstand - Volts, rms, minimum	2000
Insulation resistance - Megohms/1000 ft., minimum	500
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage	
Group B:	
Cold bending, cable - (See details for mandrel size)	No damage
Gas flame, 1 hour at 120/70 volts, rms	No failure
Physicals (unaged)	
Insulation	
Tensile strength - psi, minimum	750
Elongation - Percent, minimum	125
Jacket	
Tensile strength - psi, minimum	800
Elongation - Percent, minimum	250
Durometer (Type A) hardness, minimum (on specimen of jacket) (ASTM-D 2240)	60
Group C:	
Physicals - Cable Aging (260° ± 10°C.)	
Jacket	
Tensile strength - psi, minimum	600
Elongation - Percent, minimum	150

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical plus all of Groups A, B and C plus the following:

Cable aging and compatibility (125° ± 5°C.)	No failure
Gas flame (3 hours) at 120/70 volts, rms	No failure
Armor . . . . . Conformance to material, construction, coverage	
Specific gravity of extruded silicone insulation, maximum	1.55

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UNIT ORDER LENGTHS:

All types and sizes . . . . . 100 feet, minimum.

Custodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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